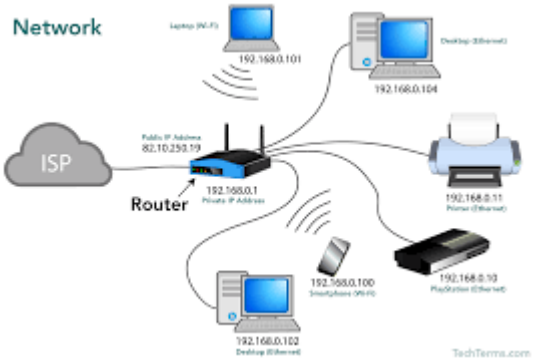
Networking Basics & Introduction to Ethical Hacking

1. What is Networking?

Networking is the process of connecting computers and devices to share data, resources, and services over wired or wireless connections.



2. Types of Networks

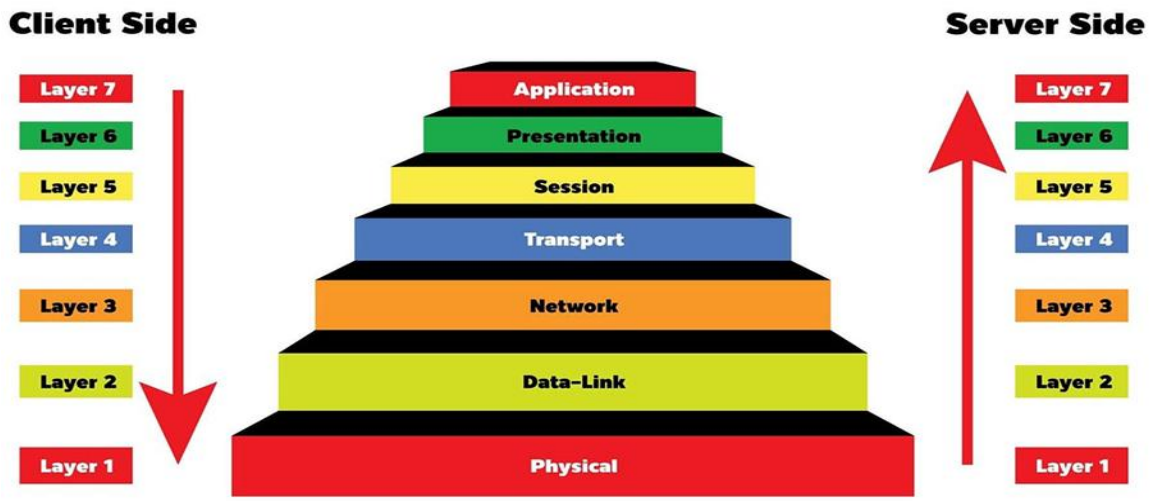
- LAN (Local Area Network)
- WAN (Wide Area Network)
- MAN (Metropolitan Area Network)
- PAN (Personal Area Network)

3. Key Networking Concepts

- IP Address: Unique identifier for devices
- DNS: Converts domain names to IP addresses
- Protocols: Rules for communication (TCP/IP, HTTP, FTP)
- Router & Switch: Devices used to connect networks

4. OSI Model (Layers)

Physical, Data Link, Network, Transport, Session, Presentation, Application



5. Introduction to Ethical Hacking

Ethical hacking involves testing systems for vulnerabilities to improve security.

6. Basic Hacking Concepts

- Vulnerability: Weakness in a system
- Exploit: Method to attack vulnerability
- Phishing: Fake emails to steal data
- Brute Force: Trying multiple passwords
- Social Engineering: Manipulating users

7. Types of Hackers

- White Hat: Ethical hackers
- Black Hat: Malicious hackers
- Grey Hat: In-between

8. Safety Tips

- Use strong passwords
- Enable two-factor authentication
- Avoid suspicious links
- Keep software updated